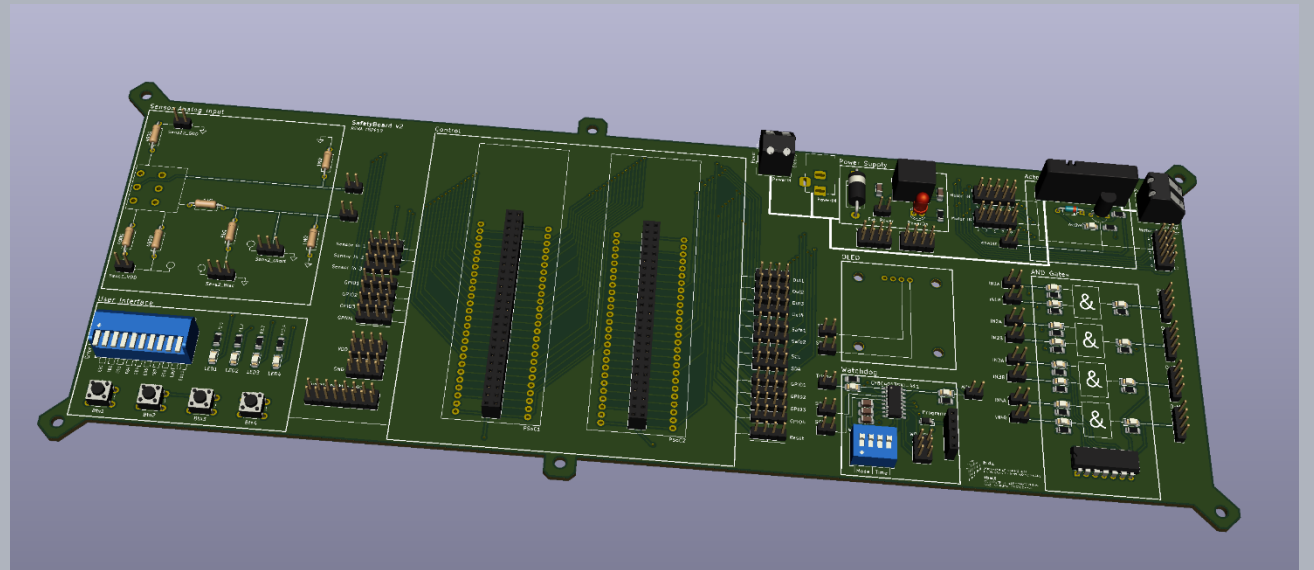


Safety in Embedded Control Systems

Organisation

Prof. Dr.-Ing. Peter Fromm



Content

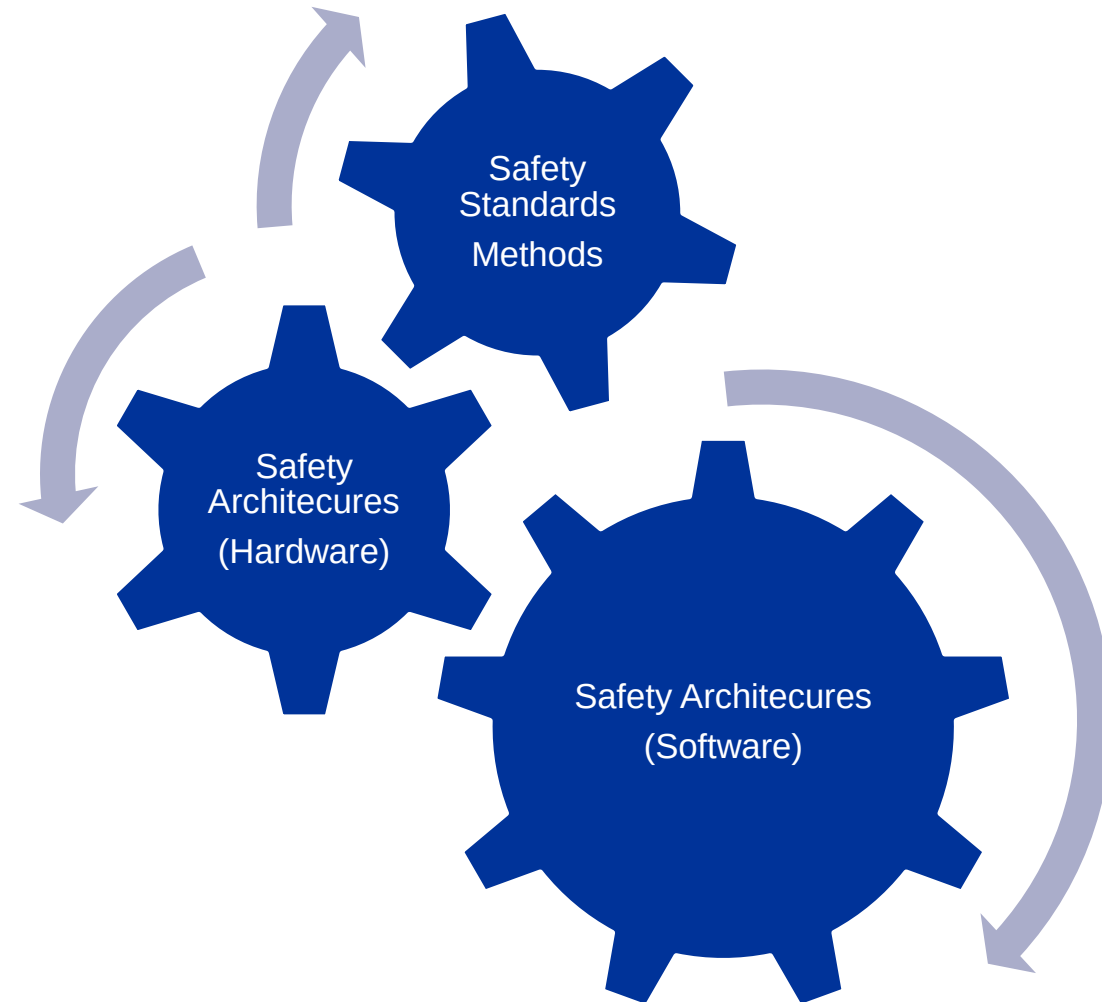
- Course Goal
- Content
- Practical Parts

Course Content

Participants will be exposed to and gain working experience with safety standards and safety architectures for embedded control systems. The course will cover

- introduction to safety standards like IEC61508 and ISO26262,
- analysis of safety cases,
- fundamental concepts for functional safety,
- development of fail safe and fail operational architectures
- concepts for avoiding systematic software errors (coding standards like MISRA, reviews, test strategies),
- concepts for dealing with sporadic errors,
- design patterns for freedom from interference,
- analysis of the features of modern safety controllers.

Big Picture



Structure of the Lecture

Week	Content
1	Introduction and Motivation
2	Safety Standards (IEC 61508, ISO26262,...)
3	Hazard and Risk Assessment, Safety Integrity Level, Safety Goals and Functions, FMEDA
4	System Level Safety Architecture - key requirements and patterns, failsafe versus fail operational
5	Hardware Architecture - reliability, microcontroller features, pin behaviour, safety manuals
6	Software Architecture - safe code, qualification, error detection and handling, freedom from interference
7	Implementation - sensor diagnostics, single controller, application level error handling, system state machine
8	Implementation - communication, system supervision (power), actor supervision, dual controller
9	Implementation - OS and system level supervision, watchdogs, alive, deadline and control flow monitoring
10	Qualification - Test and Reviews, Static Code Analysis, Metrics

Learning Outcome

to understand:

- the structure and concept of safety standards
- the development of a safety case

to apply:

- the gained knowledge to implement an embedded safety architecture (fail safe)
- systematic methods to limit systematic software errors (coding standards)

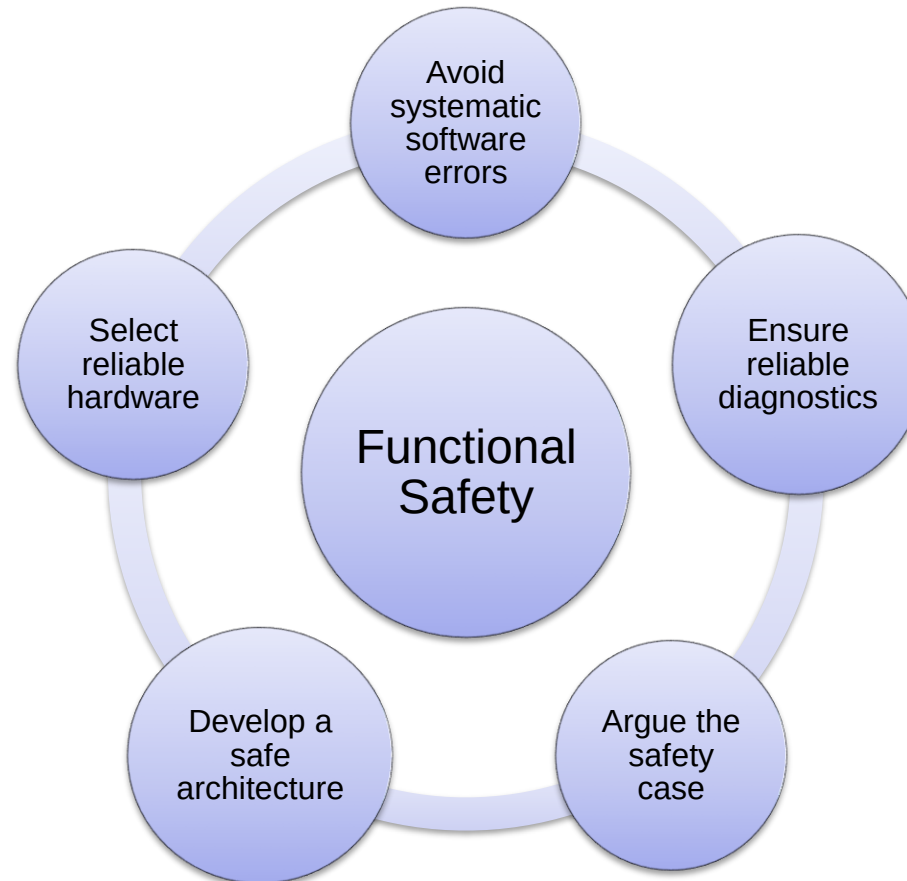
to transfer:

- the design patterns and concepts to more complex fail operational architectures.

Awareness: Be aware of **safety needs** and potential **safety risks**.
Solution: Be able to implement and assess **elementary safety functions**.

You are not a safety expert yet!

- This lecture is an introduction.
- For expert level, additional trainings and qualifications will be required.
- And you need rock solid engineering experience!



Although there is no lab part...

You are welcome to program the concepts on the LabBoard or any microcontroller of your choice.

All you need

- Microcontroller with ADC/GPIO
- Potentiometer (best case a dual one)
- Some resistors
- Some LED's

